

## DV SERIES ▪ LOW IMPEDANCE, 105°C TYPE

## KEY FEATURES

- ALUMINUM ELECTROLYTIC CAPACITOR ▪ SMD type
- Low impedance and long life



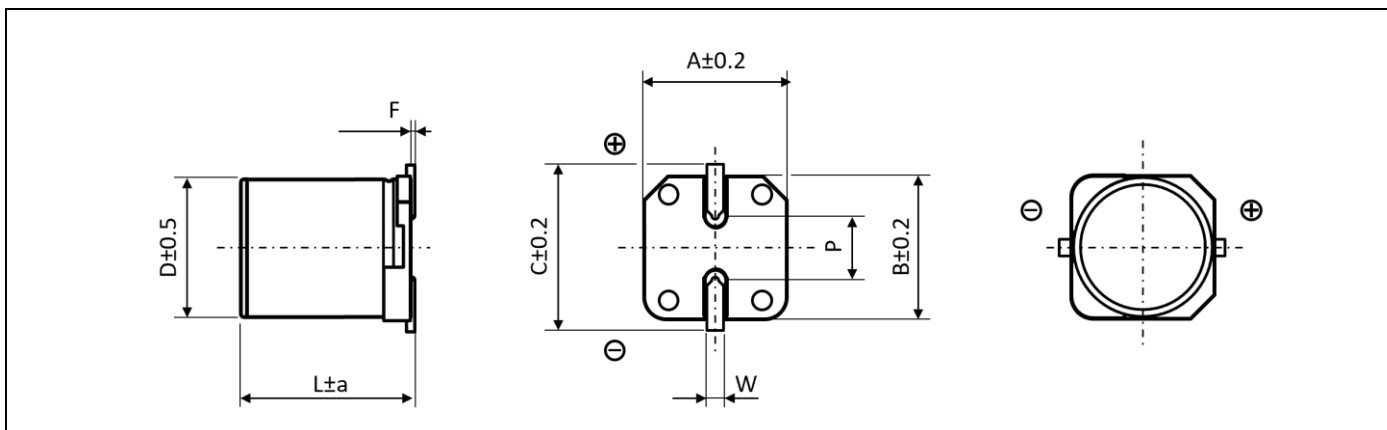
## SPECIFICATIONS

| Part Number                                                                                                                                                                                                |                   | DV101M010E055ETRZU |          | Unit          | Conditions                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------|----------|---------------|-----------------------------|
| Capacitance                                                                                                                                                                                                | $C_R$             | 100                |          | $\mu\text{F}$ |                             |
| Rated Voltage                                                                                                                                                                                              | $V_R$             | 10                 |          | V DC          |                             |
| Surge Voltage                                                                                                                                                                                              | $V_S$             | 12                 |          | V DC          |                             |
| Cap. Tolerance                                                                                                                                                                                             | $\Delta C$        | -20~20             |          | %             | 120Hz ▪ 20°C                |
| Ripple Current                                                                                                                                                                                             | $I_R$             | 156                |          | mA            | 120Hz ▪ 105°C               |
| Ripple Current                                                                                                                                                                                             | $I_R$             | 240                |          | mA            | 100KHz ▪ 105°C              |
| Dissipation Factor                                                                                                                                                                                         | $\tan\delta$      | 19                 |          | %             | 120Hz ▪ 20°C                |
| Leakage Current                                                                                                                                                                                            | $I_{\text{LEAK}}$ | 10                 |          | $\mu\text{A}$ | 2 min ▪ 20°C                |
| Max. ESR                                                                                                                                                                                                   | ESR               | 0.5                |          | $\Omega$      | 100KHz ▪ 20°C               |
| Size                                                                                                                                                                                                       | D x L             | 6.3X5.5            |          | mm            |                             |
| Operating Temperature Range                                                                                                                                                                                |                   | -55~105            |          | °C            |                             |
| AEC-Q200 Qualified                                                                                                                                                                                         |                   | NO                 |          |               |                             |
| Vibration Proof Version                                                                                                                                                                                    |                   | NO                 |          |               |                             |
| Lifetime Test                                                                                                                                                                                              |                   |                    |          |               |                             |
| Endurance<br>$V_R$ applied                                                                                                                                                                                 | Test              | 2000               |          | h             | 105°C                       |
|                                                                                                                                                                                                            | $\Delta C/C_R$    | $\leq$             | $\pm 30$ | %             | of initial measured value   |
|                                                                                                                                                                                                            | $\tan\delta$      | $\leq$             | 300      | %             | of initial specified value  |
|                                                                                                                                                                                                            | $I_{\text{LEAK}}$ | $\leq$             |          |               | the initial specified value |
| Shelf Life<br>$V_R = 0$                                                                                                                                                                                    | Test              | 1000               |          | h             | 105°C                       |
|                                                                                                                                                                                                            | $\Delta C/C_R$    | $\leq$             | $\pm 30$ | %             | of initial measured value   |
|                                                                                                                                                                                                            | $\tan\delta$      | $\leq$             | 300      | %             | of initial specified value  |
|                                                                                                                                                                                                            | $I_{\text{LEAK}}$ | $\leq$             |          |               | the initial specified value |
| Before measurement: Restore capacitor to 20°C, apply $V_R$ for 30 min according JIS-C-5101-4                                                                                                               |                   |                    |          |               |                             |
| Resistance to<br>Soldering Heat                                                                                                                                                                            | $\Delta C/C_R$    | $\leq$             | $\pm 10$ | %             | of initial measured value   |
|                                                                                                                                                                                                            | $\tan\delta$      |                    |          |               | Less than specified value   |
|                                                                                                                                                                                                            | $I_{\text{LEAK}}$ |                    |          |               | Less than specified value   |
| The capacitors shall be kept on a hot plate maintained at 250°C for 30 seconds. After removing from the hot plate and restored at room temperature, they meet the characteristic requirements listed below |                   |                    |          |               |                             |

## PART NUMBER SYSTEM

| Series | $C_R$<br>( $\mu\text{F}$ ) | $\Delta C$<br>(%) | $V_R$<br>(V) | D<br>(mm) | L<br>(mm) | Type                | Lead Treatment | Special                    |
|--------|----------------------------|-------------------|--------------|-----------|-----------|---------------------|----------------|----------------------------|
| 1_2    | 3_4_5                      | 6                 | 7_8_9        | 10        | 11_12_13  | 14                  | 15_16          | 17_18_19_20                |
| DV     | 101                        | M                 | 010          | E         | 055       | E                   | TR             | ZU                         |
|        | 100                        | -20~20            | 10           | 6.3       | 5.5       | With lead treatment | Process code   | Z=ESR or Imp.<br>U=Package |
|        |                            |                   |              |           |           |                     |                |                            |
|        |                            |                   |              |           |           |                     |                |                            |

## DIMENSIONS ■ In mm



| $\phi D$ | L   | a         | A   | B   | C   | F        | W       | P $\pm 0.2$ |
|----------|-----|-----------|-----|-----|-----|----------|---------|-------------|
| 6.3      | 5.5 | $\pm 0.2$ | 6.6 | 6.6 | 7.2 | 0.3 max. | 0.5~0.8 | 2.2         |

## MULTIPLIER $K_f$ for RIPPLE CURRENT vs. FREQUENCY

| Frequency (Hz) | 60   | 120  | 300  | 400  | 500 | 1k   | 3k   | 10k  | 50k | 100k | 300k |
|----------------|------|------|------|------|-----|------|------|------|-----|------|------|
| $K_f$          | 0.52 | 0.65 | 0.65 | 0.65 | 0.8 | 0.89 | 0.89 | 0.97 | 1   | 1    | 1    |

## PRECAUTIONS, GUIDELINES AND PACKAGING INFORMATION

Unless otherwise agreed in individual specifications, all products are subject to our “General Precautions and Guidelines” as well as our “Packaging Information”. Please refer to the following links in the table.

|                                                      |                                       |                                         |                           |                                   |
|------------------------------------------------------|---------------------------------------|-----------------------------------------|---------------------------|-----------------------------------|
|                                                      |                                       |                                         |                           |                                   |
| <a href="#">General Precautions &amp; Guidelines</a> | <a href="#">Packaging Information</a> | <a href="#">Vibration Test Profiles</a> | <a href="#">3D Models</a> | <a href="#">Reliability Tests</a> |

### DISCLAIMER

All product related data (e.g. specification, statements and general information) are subject to change without any notice. It is necessary that the customer observes all product related technical / application information and handling instructions.

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Particular operating conditions (ambient temperature, ripple current, voltage, thermal resistance, etc.) as well as storage, production or assembly may affect the performance and the lifetime of the capacitor. Please consult CapXon for lifetime estimation, failure mode considerations or worst-case scenarios according to the product technology, product tolerances / deviations or change of the characteristics of the capacitor due to shipment, storage, handling, production and usage.

For aerospace or military application, life-saving, life-sustaining, safety critical applications or any application where failure may cause severe personal injury or death, please consult us before design-in the capacitor in your application.

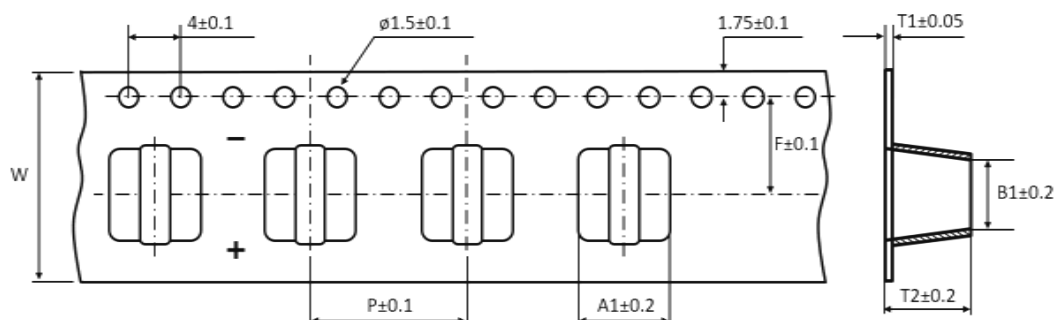
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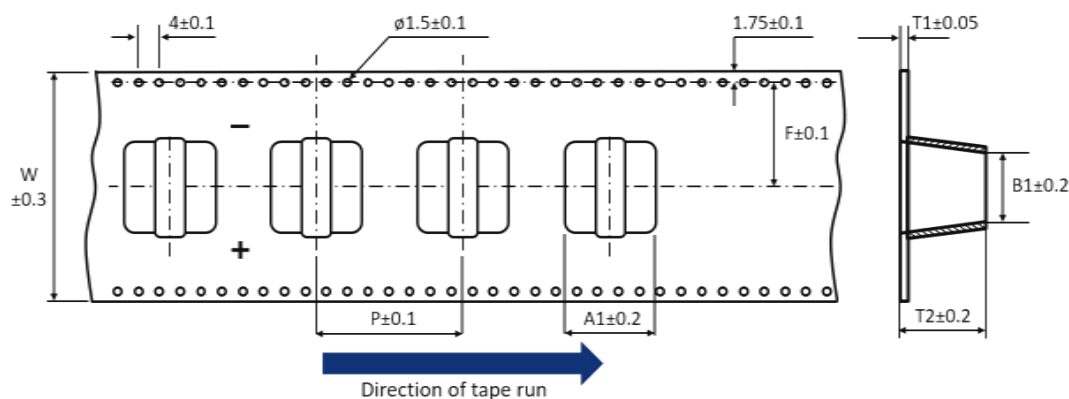
### ADDITIONAL SPECIFICATION

#### a. Tape information

SMD taping •  $\varnothing D$  4mm to 10mm



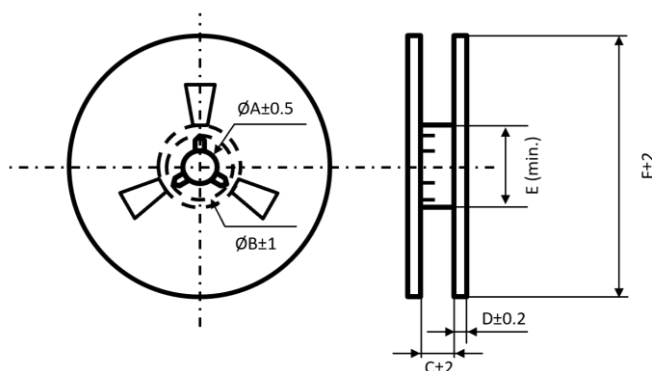
SMD taping •  $\varnothing D \geq 12.5$ mm



All dimensions in mm

| $\varnothing D \times L$ | 4 x 5.5 | 5 x 5.5 | 6.3 x 5.5 | 6.3 x 6.1 | 6.3 x 7.7 | 8 x 6.5 | 8 x 10.5 | 10 x 10.5 | 12.5 x 14 | 16 x 17 | 16 x 21.5 | 18 x 16.5 | 18 x 21.5 |
|--------------------------|---------|---------|-----------|-----------|-----------|---------|----------|-----------|-----------|---------|-----------|-----------|-----------|
| W                        | 12      | 12      | 16        | 16        | 16        | 16      | 24       | 24        | 32        | 44      | 44        | 44        | 44        |
| P                        | 8       | 12      | 12        | 12        | 12        | 12      | 16       | 16        | 24        | 28      | 32        | 32        | 32        |
| F                        | 5.5     | 5.5     | 7.5       | 7.5       | 7.5       | 7.5     | 11.5     | 11.5      | 14.2      | 20.2    | 20.2      | 20.2      | 20.2      |
| A1                       | 4.7     | 5.7     | 7         | 7         | 7         | 8.7     | 8.7      | 10.7      | 13.9      | 17.5    | 17.5      | 19.5      | 19.5      |
| B1                       | 4.7     | 5.7     | 7         | 7         | 7         | 8.7     | 8.7      | 10.7      | 13.9      | 17.5    | 17.5      | 19.5      | 19.5      |
| T1                       | 0.4     | 0.4     | 0.4       | 0.4       | 0.4       | 0.4     | 0.4      | 0.4       | 0.5       | 0.5     | 0.5       | 0.5       | 0.5       |
| T2                       | 5.7     | 5.7     | 5.7       | 6.2       | 8.1       | 7       | 11       | 11        | 14.5      | 17.5    | 23        | 17.5      | 23        |

#### b. Reel information



All reel dimensions in mm

| ∅ D | 4   | 5   | 5   | 6.3 | 6.3 | 6.3 | 6.3 | 8   | 8   | 8    | 8    | 10  | 10   | 10   | 10   | 12.5 | 16  | 16   | 18   | 18   |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|-----|------|------|------|------|-----|------|------|------|
| L   | 5.5 | 5.5 | 5.8 | 5.5 | 5.8 | 6.1 | 7.7 | 6.5 | 7.7 | 10.5 | 11.7 | 8.7 | 10.5 | 12.4 | 16.5 | 14   | 17  | 21.5 | 16.5 | 21.5 |
| A   | 15  | 15  | 15  | 15  | 15  | 15  | 15  | 15  | 15  | 15   | 15   | 15  | 15   | 15   | 15   | 15   | 15  | 15   | 15   | 15   |
| B   | 21  | 21  | 21  | 21  | 21  | 21  | 21  | 21  | 21  | 21   | 21   | 21  | 21   | 21   | 21   | 21   | 21  | 21   | 21   | 21   |
| C   | 14  | 14  | 14  | 18  | 18  | 18  | 18  | 18  | 18  | 26   | 26   | 26  | 26   | 26   | 26   | 34   | 46  | 46   | 46   | 46   |
| D   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3   | 3    | 3    | 3   | 3    | 3    | 3    | 3    | 3   | 3    | 3    | 3    |
| E   | 80  | 80  | 80  | 80  | 80  | 80  | 80  | 80  | 80  | 80   | 80   | 80  | 80   | 80   | 80   | 80   | 80  | 80   | 80   | 80   |
| F   | 380 | 380 | 380 | 380 | 380 | 380 | 380 | 380 | 380 | 380  | 380  | 380 | 380  | 380  | 380  | 380  | 380 | 380  | 380  | 380  |

Remark: Standard = Paper reel

All reel dimensions in mm

| ∅ D | 4    | 5    | 5    | 6.3  | 6.3  | 6.3  | 6.3  | 8    | 8    | 8    | 8    | 10   | 10   | 10   | 10   | 12.5 | 16   | 16   | 18   | 18   |
|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| L   | 5.5  | 5.5  | 5.8  | 5.5  | 5.8  | 6.1  | 7.7  | 6.5  | 7.7  | 10.5 | 11.7 | 8.7  | 10.5 | 12.4 | 16.5 | 14   | 17   | 21.5 | 16.5 | 21.5 |
| A   | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 | 13.2 |
| B   | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 | 21.5 |
| C   | 14   | 14   | 14   | 18   | 18   | 18   | 18   | 18   | 18   | 26   | 26   | 26   | 26   | 26   | 26   | 34   | 46   | 46   | 46   | 46   |
| D   | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  | 1.5  |
| E   | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  | 100  |
| F   | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  | 380  |

Remark: Plastic reel = Add code "U" at the end of the part number

## REVISION RECORDS

| Ver. | Date       | Process Number   | Changed by | Description           |
|------|------------|------------------|------------|-----------------------|
| 001  | 2024-10-25 | TBAKA-2410000068 | Xu XiaoJun | Initial specification |
| 002  |            |                  |            |                       |
| 003  |            |                  |            |                       |
| 004  |            |                  |            |                       |
| 005  |            |                  |            |                       |
| 006  |            |                  |            |                       |

## CUSTOMER APPROVAL

| Date | Name | Department | Signature |
|------|------|------------|-----------|
|      |      |            |           |
|      |      |            |           |
|      |      |            |           |
|      |      |            |           |